

PAPER_1494_RHO_VAC_TOTAL_877

Daniel T. Murphy

PAPER_1494 — UQFF: PAPER_877 ρ_{vac} Total = $11 \cdot \rho_{\text{SCm}}$ (Bucket C)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket C — extracted from PAPER_877 Three-Assumption Cosmogenesis

Date: June 17, 2026

Location: 41.0997 N, 80.6495 W (Youngstown, OH, USA)

Status: CLOSED — $\rho_{\text{vac_total}} = (N_{\text{CH}} + 2) \times \rho_{\text{SCm}} = 11 \times \rho_{\text{SCm}} = 7.799 \times 10^{30} \text{ J/m}^3$ EXACT

Observation

PAPER_877 Three-Assumption Cosmogenesis defines $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}}$. With canonical $\rho_{\text{UA}} = 10 \cdot \rho_{\text{SCm}}$, $\rho_{\text{vac}} = 11 \cdot \rho_{\text{SCm}} = 7.799 \times 10^{30} \text{ J/m}^3$.

UQFF Closed Identity

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$\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 10 \cdot \rho_{\text{SCm}} + \rho_{\text{SCm}} = 11 \cdot \rho_{\text{SCm}} = 11 \times 7.09 \times 10^{29} = 7.799 \times 10^{30} \text{ J/m}^3$ EXACT

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Physical Interpretation

The total vacuum energy density is the sum of UA (universal aether, 10×) and SCm (superconducting matter, 1×) layers. Provides the structural baseline for the 3-assumption cosmogenesis model in PAPER_877.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model frameworks describe vacuum and atomic structure via different mechanisms. Closure uses only canonical UQFF integer primitives $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$ and $\rho_{\text{SCm}} = 7.09 \times 10^{29} \text{ J/m}^3$. **Zero free parameters.**

Reference

- Source paper: PAPER_877 Three-Assumption UQFF Cosmogenesis Master Equation (Daniel T. Murphy, April 7, 2026)
- Related: PAPER_872 (proto-element Z identity), PAPER_646 ($F_{U_{Bi_j}}$ / DPM mechanism), PAPER_062 (DPM vacuum manifold), PAPER_1051 (two-component ρ aether).

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